Category

has
- objects
- morphisms

statisfies
- composition
- asociativity
- identity

Terminal object 1

$\forall A \in Ob\,\mathcal{C}.\ \exists!\ !_A : A \to 1.$

Examples of Categories

- **Sets**
  - A • sets
  - $\to$ • functions between sets
- **$Sets_{fin}$**
  - A • finite sets
  - $\to$ • functions between sets
- **poset**
  - A • partialy ordered set with binary relation $a \leq_A b$ such that we have
    - reflexivity
    - trasitivity
    - antisymetry
  - $\to$ • monotone funkcije

- **Rel** A• sets
  →• binary relations
  $(R : A \longrightarrow B = \{(a,b) \in A \times B ;\ a \leq_R b\})$
  - composition:
    $S \circ R = \{(a,c) \in A \times C ;\ \exists b \in B.\ (a,b) \in R, (b,c) \in S\}$
- **preorder** A• elements of P (preordered set)
  →• relation $(a \to b = a \leq b)$

  Relation has to be
  - reflexive
  - transitive
- **Monoid** A• * (one object)
  →• elements of monoid M
  - $id_* = 1$
  - composition: $x \circ y = x \cdot y$
- $\mathbb{C}^{op}$ (dual)

  A• objects of $\mathbb{C}$ $(A \rightsquigarrow A^*)$
  →• $B^* \xrightarrow{f^*} A^* \iff A \xrightarrow{f} B$
  - $f^* \circ g^* = (g \circ f)^*$

- $\mathbb{C}^{\to}$ (arrow category)

  A• arrows of $\mathbb{C}$

  →• $g: f \longrightarrow f'$ is a pair (g_1, g_2) such that diagram commutes

$$\begin{array}{ccc} A & \xrightarrow{g_1} & A' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ B & \xrightarrow[g_2]{} & B' \end{array} \qquad (g_2 \circ f = f' \circ g_1)$$

- $\mathbb{C}/C$ (slice category)

  A• arrows $A \longrightarrow C$

  →• $a: f \longrightarrow g$ (where $f: A \to C$, $g: B \to C$) is an arrow $a: A \to B$ such that $g \circ a = f$

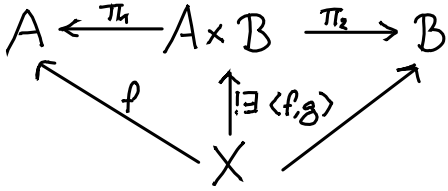
$$\begin{array}{ccc} A & \xrightarrow{a} & B \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle g} & \\ & C & \end{array}$$

Construction on categories

- **Product** $A \times B$

The product of two objectct A and B is

1. An object $A \times B$
2. Two projection morphisms $\pi_1 : A \times B \to A$, $\pi_2 : A \times B \to B$, such that $\forall X \in Obj.\ \forall f : X \to A\ .\ \forall g : X \to B.$
$!\exists \langle f,g \rangle : X \to A \times B\ .\ \pi_1 \circ \langle f,g \rangle = f \wedge \pi_2 \circ \langle f,g \rangle = g$



Category with finate products has

1. Terminal object
2. Binary product $A \times B$ for every pair A, B

Properties

- uniqueness up to isomorphism
(There exist many products, but they are isomorphic)
- $A \times B \cong B \times A$
- $(A \times B) \times C = A \times (B \times C)$
- $A \times 1 = A$

$f : A \longrightarrow B$

monomorphism: $\forall g,h : C \to A \,.\, fg = fh \Rightarrow g = h$

epimorphism: $\forall g,h : C \to A \,.\, gf = hf \Rightarrow g = h$

split mono: arrow with right inverse

split epi: arrow with left inverse

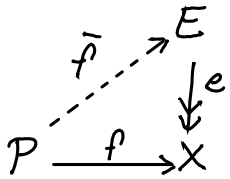
$es = 1_A$ $\quad$ e... section

$\quad\quad\quad\quad\quad\quad$ s... retraction

Projective object

P projective if

$\forall e \text{ epi}: E \twoheadrightarrow X \,.\, \forall f\, P \to X \,.\, \exists \bar{f} : P \to E \,.\, e \circ \bar{f} = f$



Initial object 0. $\quad \forall C \in \mathcal{C} \,.\, \exists!\, 0 \to C$

terminal object 1. $\quad \forall C \in \mathcal{C} \,.\, \exists!\, C \to 1$

Generalized elements of A: any arrow $x : X \to A$ from any X.

Homesets (on locally small categories)

$Hom(A,B) = \{ f \in \mathcal{C} \,;\, f : A \to B \}$

$Hom(A,g) : Hom(A,B) \longrightarrow Hom(A,B')$

$f \mapsto gf$

$Hom(X, A \times B) = Hom(X,A) \times Hom(X,B)$